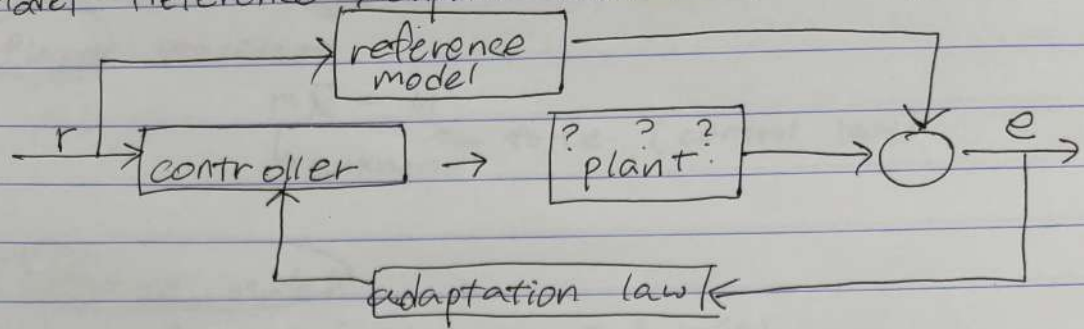


① Model-Reference Adaptive Control (MRAC).



- plant : structure known + parameters unknown.
- reference model : ideal response to external command.
- controller : input to plant.
 $\Rightarrow$ tracking convergence. ($e \rightarrow 0$)
- adaptation : ~~add~~ update of control.

example 8.1.

Plant structure

$$m\ddot{x} = u.$$

τ unknown force. (control law)

reference model

$$\ddot{x}_m + \lambda_1 \dot{x}_m + \lambda_2 x_m = \lambda_2 r(t).$$

i) If m is known, u is simply,

$$u = m(\ddot{x}_m - 2\lambda\dot{\tilde{x}} - \lambda^2\tilde{x}) \quad (\text{control law})$$

$$\text{for } \tilde{x} = x - x_m.$$

to get

$$\ddot{\tilde{x}} + 2\lambda\dot{\tilde{x}} + \lambda^2\tilde{x} = \left(\frac{d}{dt} + \lambda\right)^2 \tilde{x} = 0.$$

$$\Rightarrow \tilde{x} \rightarrow 0.$$

ii) Assume m is not known.

We use

$$u = \hat{m}(t) \cdot (\ddot{x}_m - 2\lambda\dot{\tilde{x}} - \lambda^2\tilde{x}).$$

τ adjustable parameter.

to find

$$\begin{aligned} m(\ddot{\tilde{x}} + 2\lambda\dot{\tilde{x}} + \lambda^2\tilde{x}) &= m(\ddot{x} - \ddot{x}_m + 2\lambda\dot{\tilde{x}} + \lambda^2\tilde{x}) \\ &= \hat{m}(\ddot{x}_m - 2\lambda\dot{\tilde{x}} - \lambda^2\tilde{x}) + m(-\ddot{x}_m + 2\lambda\dot{\tilde{x}} + \lambda^2\tilde{x}) \\ &= (\hat{m} - m)\ddot{x}_m - 2(\hat{m} - m)\lambda\dot{\tilde{x}} - (\hat{m} - m)\lambda^2\tilde{x} \\ &= (\hat{m} - m)(\ddot{x}_m - 2\lambda\dot{\tilde{x}} - \lambda^2\tilde{x}) \end{aligned}$$

By setting $s := \dot{\tilde{x}} + \lambda\tilde{x}$,

$$\tilde{m} := \hat{m} - m, \quad v := \ddot{x}_m - 2\lambda\dot{\tilde{x}} - \lambda^2\tilde{x}.$$

to rewrite

$$m(\dot{s} + \lambda s) = \tilde{m}v.$$

We set $\dot{\hat{m}} = -\gamma v s$. (adaptation law)

adaptation gain.

Stability (bdd) & Convergence ?

→ use Lyapunov theory.

Set Lyapunov ft

$$V := \frac{1}{2} \left[m s^2 + \frac{1}{\gamma} \tilde{m}^2 \right].$$

$$\begin{aligned} \text{Then, } \dot{V} &= m \dot{s} s + \frac{1}{\gamma} \tilde{m} \dot{\tilde{m}} \\ &= (\tilde{m} \gamma - m \lambda s) s + \frac{1}{\gamma} (-\gamma \gamma s) \tilde{m} \\ &= \tilde{m} \gamma s - m \lambda s^2 - \gamma s \tilde{m} = -m \lambda s^2 \leq 0. \end{aligned}$$

Since $\dot{V} \leq 0$ and $V \geq 0$, V is bounded.

So, s & $\tilde{m}$: bdd. V converges.

$\Rightarrow \dot{V}$: uniformly continuous.

< Barbalat's Lemma >

If the diff'ble ft $f(t)$ has a finite limit and $\dot{f}$ is uniformly continuous, then $\dot{f} \rightarrow 0$.

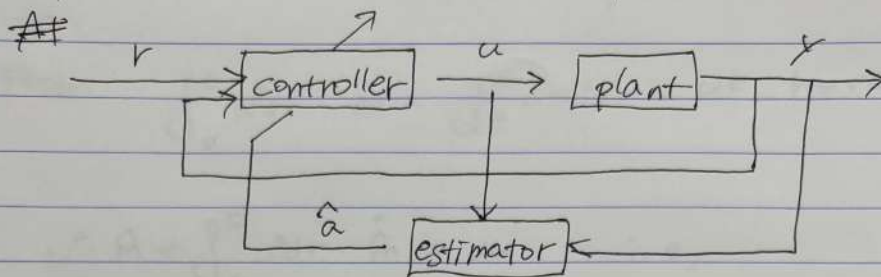
By Barbalat's lemma,

$$\dot{V} \rightarrow 0, \text{ i.e., } s \rightarrow 0.$$

$$\therefore \tilde{x}, \tilde{\dot{x}} \rightarrow 0. \text{ (tracking error)}$$

② Self-Tuning Controllers (STC)

⇒ Estimate the unknown parameters.



$\hat{a}$: estimated parameters.

Example 8.2

$$\hat{m}(t) = \frac{u(t)}{\ddot{x}} \rightarrow \text{not good.}$$

i) measurement of $\ddot{x}$ could be inaccurate.

ii) $\ddot{x}$ can be close to zero

Instead, we use L-S. (least squares) approach.

i.e., find $\hat{m}(t)$ s.t. for $e(t) = \hat{m}\ddot{x}(t) - u(t)$,

$J = \int_0^t e^2(t) dt$ is minimal. prediction error.

Differentiate (write $\ddot{x} = w$).

$$\int_0^t e^2(t) dt = \int_0^t (\hat{m}^2 w^2 - 2\hat{m}w \cdot u + u^2) dt$$

w.r.t. $\hat{m}$, to get

$$\hat{m} = \frac{\int_0^t w u dt}{\int_0^t w^2 dt}$$

- If m is slowly time-varying, the above estimate has to be recalculated at every new time instant.

From $\int_0^t w^2 dr \cdot \hat{m} = \int_0^t w u dr$, we have

$$w^2 \cdot \hat{m} + \int_0^t w^2 dr \cdot \dot{\hat{m}} = w u \quad \text{i.e.,}$$

$$\dot{\hat{m}} = \frac{w(u - w\hat{m})}{\int_0^t w^2 dr} = \frac{-we}{\int_0^t w^2 dr}$$

Using the notation

$$P(t) := \frac{1}{\int_0^t w^2 dr},$$

it can be rewritten as

$$\dot{\hat{m}} = -Pwe \quad \dots \textcircled{A}$$

Also, note that $\frac{d}{dt}\left(\frac{1}{P}\right) = w^2 \quad \dots \textcircled{B}$

We can implement this method by using $\textcircled{A}$ $\textcircled{B}$.

Lem 8.1. Consider two signals $e \in \Phi$ related by

$$e(t) = H(p) [k \Phi^T(t) \cdot v]$$

where $H(p)$: strictly positive real transfer fct,

$\text{sgn}(k)$: known (k : const).

$v(t)$: mble vector.

If $\dot{\Phi} = -\text{sgn}(k) \cdot \gamma \cdot e v(t) \quad (\gamma > 0)$,

then $e(t), \Phi(t)$: globally bdd.

Furthermore, if v is bdd, then $e(t) \rightarrow 0$

8.2.

Adaptive Control of First-Order Systems.

Plant : $\dot{y} = -\underbrace{a_p}_{\substack{\text{consts.} \\ \text{(unknown)}}} y + \underbrace{b_p}_{\text{plant input}} u$.

plant output

Reference model : $\dot{y}_m = -\underbrace{a_m}_{\substack{\text{consts} \\ \text{(known)}}} y_m + \underbrace{b_m}_{\substack{\text{external reference} \\ \text{signal (bdd)}}} r(t)$.

$y_m = M r$, for a transfer M .

$M = \frac{b_m}{p + a_m}$, p : Laplace variable.
 $\hookrightarrow$ SPR ft.

Goal : Formulate a control law & adaptation law.

s.t. $y(t) - y_m \rightarrow 0$ as $t \rightarrow \infty$.

assumption : $\text{sign}(b)$ is known.

> Choice of control law.

Control law : $u = \hat{a}_r r + \hat{a}_y y$,

$\hat{a}_r, \hat{a}_y$: variable feedback gains.

$$\begin{aligned} \dot{y} &= -a_p y + b_p (\hat{a}_r r + \hat{a}_y y) \\ &= -(a_p - \hat{a}_r b_p) y + \hat{a}_r b_p r. \end{aligned}$$

If the plant parameters were known,

$$a_r^* = \frac{b_m}{b_p}, \quad a_y^* = \frac{a_p - a_m}{b_p}.$$

would lead to the closed-loop dynamics

$$\dot{y} = -a_m y + b_m r \Rightarrow \text{error} \rightarrow 0.$$

Now, we assume a_p & b_p are unknown.

> Choice of adaptation law.

Let $e := y - y_m$ (tracking error).

$$\tilde{a}(t) := \begin{bmatrix} \tilde{a}_r \\ \tilde{a}_y \end{bmatrix} := \begin{bmatrix} \hat{a}_r - a_r^* \\ \hat{a}_y - a_y^* \end{bmatrix}. \quad (\text{parameter errors})$$

Note that

$$\begin{aligned} \dot{e} &= \dot{y} - \dot{y}_m \\ &= -(a_p - \hat{a}_y b_p) y + \hat{a}_r b_p r + a_m y_m - b_m r \\ &= -a_m (y - y_m) + (a_m - a_p + b_p \hat{a}_y) y \\ &\quad + (b_p \hat{a}_r - b_m) r \\ &= -a_m e + b_p (\tilde{a}_y y + \tilde{a}_r r). \end{aligned}$$

$$\Rightarrow e = \frac{1}{a_r^*} M (\tilde{a}_r r + \tilde{a}_y y).$$

$$\text{with transfer } \mu = \frac{b_p}{p + a_m}.$$

Letting $\Phi = \begin{bmatrix} \tilde{a}_r \\ \tilde{a}_y \end{bmatrix}$, $v = \begin{bmatrix} r \\ y \end{bmatrix}$, Lem 8.1 suggests

$$\begin{aligned} \dot{\hat{a}}_r &= -\text{sgn}(b_p) r e r \\ \dot{\hat{a}}_y &= -\text{sgn}(b_p) r e y \end{aligned}$$

> Tracking Convergence Analysis.

Set a Lyapunov Φ

$$V(e, \Phi) := \frac{1}{2} e^2 + \frac{1}{2\delta} |b_p| (\tilde{a}_r^2 + \tilde{a}_y^2).$$

$$\begin{aligned} \dot{V} &= e \cdot \dot{e} + \frac{1}{\delta} |b_p| (\tilde{a}_r \dot{\tilde{a}}_r + \tilde{a}_y \dot{\tilde{a}}_y) \\ &= e(-a_m e + b_p (\tilde{a}_y y + \tilde{a}_r r)) - b_p \cdot e (r \tilde{a}_r + y \tilde{a}_y) \\ &= -a_m e^2 \leq 0. \end{aligned}$$

By Barbalat's lemma, we have $e \rightarrow 0$.

7

7 Parameter Convergence Analysis.

$$\text{In } e = \frac{1}{a_r^*} M(\tilde{a}_r r + \tilde{a}_y y)$$

$\downarrow$ conv. zero $\downarrow$ unif'dly conti.
 $\Downarrow$ conv. zero.

From $\dot{\hat{a}}$ & $\dot{\hat{y}}$, we have $\dot{\hat{a}}_r \rightarrow 0$, $\dot{\hat{a}}_y \rightarrow 0$.

So, at large time t , $\tilde{a} \approx \text{const}$.

$$r(t) \cdot \tilde{a}_r + y(t) \tilde{a}_y = 0.$$

Simulation

plant : $\dot{y} = y + 3u$.

ref. model : $\dot{x}_m = -2x_m + 4r$.

Simulation 1) $r(t) = 4$.

$\Rightarrow$ tracking error $e \rightarrow 0$.

parameter error $\tilde{a} \not\rightarrow 0$.

Simulation 2) $r(t) = 4 \sin(3t)$.

$\Rightarrow e \rightarrow 0$ & $\tilde{a} \rightarrow 0$.